

A Guyton-based hypothesis for venous congestion

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ABSTRACT

Despite much research and opinion, the underpinnings of venous congestion remain elusive. Many posit that elevated right atrial pressure (P_{RA}) is the *primum movens* of venous congestion. However, a recent Geometrical model (GM) built from Guyton's framework advances that P_{RA} is a *dependent* variable. The GM predicts that: 1.) venous Doppler patterns do not reflect volume state or responsiveness; 2.) venous congestion is not caused by P_{RA} and, in fact, may occur with a low or normal P_{RA} ; and 3.) that venous Doppler patterns correspond to transmural rather than absolute P_{RA} .

Key words: Guyton's cardiovascular model, venous congestion, venous excess ultrasound score (VExUS), venous Doppler, fluid responsiveness, fluid tolerance



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Background

Assessing fluid tolerance (F_{TOL}) via Doppler ultrasound is of growing clinical interest (1, 2), with significant research and debate regarding the relationship between F_{TOL} and fluid responsiveness (F_R) (3-17). The Venous Excess Ultrasound Score (VExUS) (18, 19) assesses venous congestion associated with kidney dysfunction (18) and various hemodynamic variables (5-7, 11, 12, 20). Existing explanations (4, 13-17, 21) lack a comprehensive accounting of venous congestion relative to Guyton's hemodynamic principles. We explore how a validated model of global circulation sets the boundaries within which a model of venous congestion must operate.

A recent Geometrical model (GM) (22) posits that right atrial pressure (P_{RA}) and blood flow ($Q_{CIRCULATORY}$) are both *dependent* variables, concurrently determined by interacting independent variables (Figure 1). We conclude the following: 1.) venous Doppler cannot directly measure volume state or F_R ; 2.) the root cause of venous congestion is to be found either among the independent circulatory variables or caused by local outflow restriction, but not by P_{RA} , and 3.) venous Doppler patterns reflect transmural rather than absolute P_{RA} .

Venous Doppler, volume state and fluid responsiveness

Neither venous Doppler nor P_{RA} reliably inform the clinician about volume state or fluid responsiveness, as underscored recently (5).

Venous Doppler patterns do not convey information regarding volume state

Stressed volume is arguably the best representation of 'volume state'. A validated approximation of stressed volume is the mean systemic pressure analogue (P_{MSA}) (23, 24), which combines the stressed blood volume and systemic circulation compliance (25).

$$P_{MSA} = (0.96 \times P_{RA}) + (0.04 \times MAP) + (c \times CO) \quad (1)$$

In euvoemia, the P_{MSA} is normally 8-10 mmHg (25). Considering recent data (5) we note how volume state and venous Doppler measures diverge. Calculated P_{MSA} (assuming body surface area of 1.8 m² and $c = 0.6$) was ≈ 13 mmHg for F_R patients, and ≈ 17 mmHg for fluid unresponsive patients (F_{UR}), suggesting responders were relatively volume-deficient at baseline (5). Despite this P_{MSA} (i.e., volume state) difference, baseline venous Doppler measures were nearly identical. Post-bolus, P_{MSA} increased similarly in both groups, yet Doppler measures diverged, showing greater congestion in F_{UR} patients. Thus, venous Doppler failed to differentiate volume state at baseline and post-bolus, reaffirming that great vein pressure does not correlate with volume state (26).

Venous Doppler patterns do not convey information regarding fluid responsiveness

While F_R and abnormal (i.e., 'congested') venous Doppler patterns can coexist (5, 7, 27, 28), normal venous Doppler is also common in F_{UR} patients (4, 5, 7, 27, 28). Approximately 30% of patients with normal venous Doppler and 40% with $P_{RA} < 8$ mmHg are F_{UR} (5, 7, 27-30), supporting recommendations against fluid expansion based solely on static $P_{RA} < 8$ mmHg (31). Even with low preload on ultrasound, dynamic measures are prudent.

Venous congestion and the geometrical model

The recently proposed GM is founded on Guyton's approach to the circulation, that "... *right atrial pressure is not one of the primary determinants of cardiac output but, instead, is itself determined simultaneously along with cardiac output*" (13, 17, 22, 32). The GM can help explain why venous Doppler patterns predict neither volume state nor responsiveness (13, 22). Additionally, the GM suggests that venous congestion is best understood as impaired organ function secondary to reduced venous drainage caused by the independent circulatory variables, or local critical collapse pressure (P_{CRIT}). Venous congestion, therefore, may be

associated with elevated absolute or transmural P_{RA} or with normal P_{RA} .

Venous Doppler, volume state and responsiveness

The GM maintains that venous Doppler morphology, like P_{RA} , is a *dependent* variable. The *independent* variables determining venous Doppler morphology are shown in Figure 1. The four principle independent variables of the GM are: P_{MSA} , R_{VR} , the pericardial pressure (P_{PC}) and cardiac resistance ($R_{CARDIAC}$). Importantly, how these independent variables determine P_{RA} is mediated by the ratio of R_{VR} to $R_{CARDIAC}$, and if the patient is F_R or F_{UR} (13). Furthermore, the GM posits that the venous Doppler pattern within the VExUS score is determined by the transmural P_{RA} (i.e., $P_{RA} - P_{PC}$); the GM equations shown in Figure 1 use VExUS and $P_{RA} - P_{PC}$ interchangeably. Within the GM, the right atrial transmural pressure is set, generally, by the interaction between the P_{MSA} , the R_{VR} and $R_{CARDIAC}$ as follows, for F_R and F_{UR} patients, respectively (13, 22):

$$VExUS_{F_R} \approx \frac{P_{MSA}}{\frac{R_{VR}}{R_{CARDIAC}} + 1} \quad (2)$$

$$VExUS_{F_{UR}} \approx P_{MSA} - \frac{R_{VR}}{R_{CARDIAC}} P_{RA_{plat}} \quad (3)$$

Although we suggest this algebraic expression as an estimate of the pressure behind venous distention, it cannot be used to calculate the score itself. VExUS is an ordinal score based on qualitative waveform patterns across multiple venous territories. The equations above illustrate that the VExUS venous Doppler morphologies are directly related to P_{MSA} but indirectly related to the $R_{VR}:R_{CARDIAC}$ ratio. Increasing P_{MSA} raises VExUS, but increased R_{VR} (e.g., by increasing alpha-agonism or shunting flow through slow time-constant vascular beds) and/or decreased $R_{CARDIAC}$ (e.g. increased inotropy) diminishes VExUS for any given volume state. The $R_{VR}:R_{CARDIAC}$ ratio is analogous to ventriculo-arterial coupling (i.e., the $E_A:E_{ES}$ ratio) such that the VExUS- P_{MSA} relationship must be understood in terms of ‘veno-cardiac coupling’.

Second, in F_R patients, the $R_{VR}:R_{CARDIAC}$ ratio is the multiplicative inverse of VExUS while in F_{UR} patients the ratio is the arithmetic inverse. Thus, if volume expansion increases P_{MSA} , VExUS will rise *more* in F_{UR} patients because the venous return curve shifts right along a flat portion of the cardiac function curve (4, 5, 13). Nevertheless, this assumes equal P_{MSA} increase between F_R and F_{UR} patients. Venous Doppler morphology might, therefore, increase equally in F_R and F_{UR} patients if P_{MSA} changes differentially. For example, Ruste and colleagues observed that F_R patients had greater increases in cardiac output, P_{RA} and MAP (i.e., the determinants of the P_{MSA}) compared to F_{UR} patients (8). Accordingly, while venous Doppler will not ‘congest’ as much in F_R patients, this could be offset by greater rise in P_{MSA} .

Third, the GM accounts for the seemingly paradoxical coexistence of low VExUS in a F_{UR} patient and high VExUS in a F_R patient. In the model, the transition between F_R and F_{UR} is defined by the P_{RA} at which the cardiac function curve flattens ($P_{RA-plat}$) (Equation 3) (22). The $P_{RA-plat}$ is the P_{RA} at which the end-diastolic pressure-volume relationship inflects, i.e. when additional preload raises P_{RA} with little increase in end-diastolic volume (33). With $P_{RA} < P_{RA-plat}$, the patient is F_R and with $P_{RA} > P_{RA-plat}$, the patient is F_{UR} . Accordingly, if there is a relatively high value of $P_{RA-plat}$, (e.g., ventricular stiffening or thickened wall relative to radius (33)) the patient can be F_R even with a relatively high P_{RA} . On the other hand, when $P_{RA-plat}$ is low, F_{UR} can occur also at low P_{RA} .

Venous congestion: three hypothetical types

From the perspective of the GM, elevated P_{RA} and VExUS are both epiphenomena of congestion. However, as P_{RA} is a dependent variable, it cannot be causative. The perception that P_{RA} (or VExUS) causes organ congestion is a *tangible bias* – when a measurable entity is mistaken for root cause rather than an associated observation. The GM predicts that venous congestion is caused by the variables on the right side of the Figure 1 equations. Understanding the causative factors is critical for appropriate therapy and predicts that venous congestion may occur in three general forms; 1.) with elevated transmural P_{RA} ; 2.) with elevated absolute P_{RA} , and 3.) without P_{RA} elevation.

It is likely that venous Doppler helps differentiate elevated transmural P_{RA} (i.e., $P_{RA} - P_{PC}$) from elevated absolute P_{RA} . Early investigation found that increasing transmural P_{RA} by reducing the P_{PC} by suction led to congested venous Doppler patterns (i.e., reduced the S relative to D wave) (34). If venous Doppler morphology reflects transmural P_{RA} , then venous Doppler patterns better evaluate $R_{CARDIAC}$, especially in conjunction with a measure of flow because the transmural P_{RA} is the base of the $R_{CARDIAC}$ triangle (22). This hypothesis supports lumped Doppler measures such as the left ventricular outflow tract velocity time integral-to-VExUS ratio (35), echoes data relating depressed left and right cardiac function to VExUS (7, 10), and the opinion of the VExUS originators who believe that VExUS reflects right ventricular to pulmonary artery coupling (16).

If there is elevated P_{RA} but normal or S-wave dominant venous Doppler morphology, this suggests augmented P_{PC} as a cause of congestion (Figure 2B). The pathophysiology then is right atrial external compression (e.g., cardiac tamponade or high airway pressure expressed in the pericardial space). We note that this type of congestion would likely present with a fixed, large (>2 cm) inferior vena cava diameter, conferring a VExUS score of at least 1, however, we suggest that right atrial compression either blunts normal P_{RA} pressure transduction to the great veins or reveals $S > D$ wave because of γ -descent loss (36). Possible treatment for this type of congestion is reduction of intrathoracic pressure caused by a stiffened chest wall or extrapulmonary ARDS coupled with high airway pressure, or pericardial drain, depending on the cause.

Finally, if there is elevated P_{CRIT} the GM predicts venous hypertension *without* P_{RA} or VExUS elevation. Rising P_{CRIT} predicts both reduced P_{RA} and blood flow. This is shown in the GM equations and illustrated in Figure 2C, as recently discussed (37, 38). A clinical example would be an abdominal lymphoma with inferior vena cava and renal vein compression. This increases the P_{CRIT} , reduces venous return/total blood flow, and diminishes P_{RA} while congesting the kidneys; this highlights how neither P_{RA} nor venous

Doppler *cause* venous hypertension – in this case P_{CRIT} is the limiting independent variable. Treatment is to reduce the tumor burden and decrease P_{CRIT} (Figure 2C). In this instance, P_{RA} and venous Doppler have reduced sensitivity for detecting venous congestion because they are not physiological drivers but mere passengers.

Limitations and conclusions

First, the GM treats venous return as a lumped parameter overlooking organ-specific differences (37, 38) or distinctions between upper and lower body venous return (i.e., via the SVC and IVC, respectively) (39). Thus, the upper body could be ‘venous limited’ from an SVC Starling resistor while the lower body is unlimited; therefore, the internal jugular veins may provide additional congestion assessment data (40, 41). Second, this hypothesis takes no position on the primary histopathology of organ-specific congestion and edema which can be distinct from macrohemodynamic variables. The GM model only speaks to elevated venous pressure and not organ or tissue congestion, more broadly. This is to say that tissue edema and congestion can occur *without* venous hypertension as described by the GM above. If we consider disease states with significant capillary leak (e.g., burn injury, Clarkson syndrome) that may present with diminished stressed venous volume (i.e., reduced P_{MSA}) despite increased interstitial volume, we emphasize that Guytonian physiology and the GM might be misinterpreted and lead to inappropriate fluid administration.

By connecting venous return (P_{MSF} , R_{VR}) and cardiac function (P_{PC} , $R_{CARDIAC}$) the GM provides a global framework within which venous Doppler patterns should be interpreted. These patterns relate directly to volume state (P_{MSF} or P_{MSA}) but indirectly to the $R_{VR}:R_{CARDIAC}$ ratio, especially in F_{UR} patients. Discriminating dependent from independent hemodynamic variables is crucial for diagnosis and therapy; neither P_{RA} nor VExUS cause venous congestion, instead they are hemodynamic intermediaries, imperfect reflections of the true sources of venous hypertension.

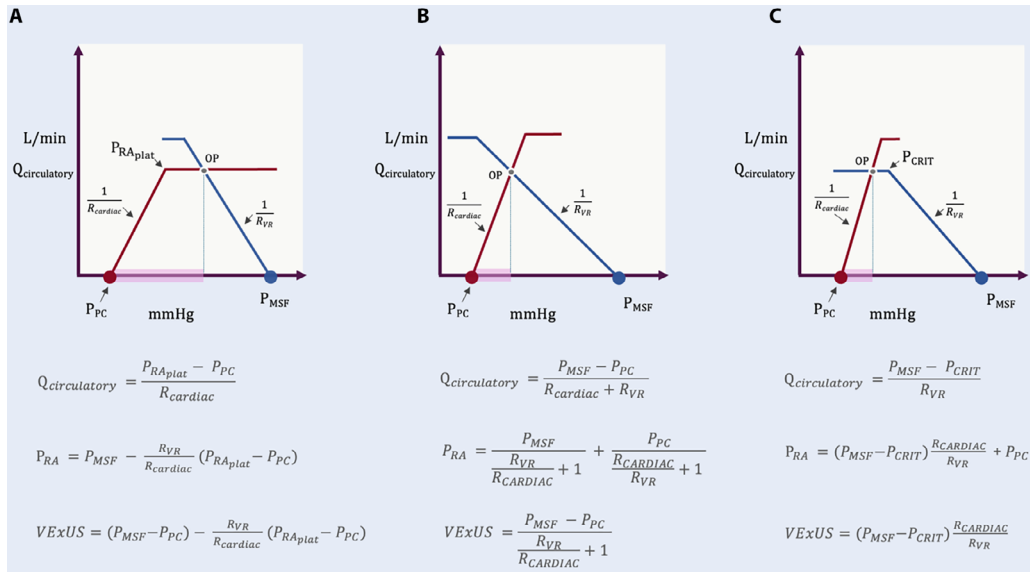


Figure 1. The Geometrical Model. This figure shows the detailed equations that predict blood flow, P_{RA} and the right atrial distending venous pressure behind VExUS (P_{RA} minus P_{PC} , see text). In the text, to simplify the discussion, Equations 2 and 3 assume that the pericardial pressure (P_{PC}) is zero; expanded equations show the relationship with VExUS when P_{PC} is included. The pink bars on the x-axis represent the VExUS distending pressure or transmural P_{RA} for each state. The intersection of cardiac function (red) and venous return curve (blue) forms the operating point (OP) which defines the P_{RA} and total blood flow (Q). The GM suggests that the circulation can exist in one of three states that are not mutually exclusive. A.) cardiac limited (or F_{UR}); B.) unlimited (F_R); C.) venous limited or venous Starling resistor physiology. P_{MSF} is mean systemic filling pressure and for discussion purposes considered the same as P_{MSA} .

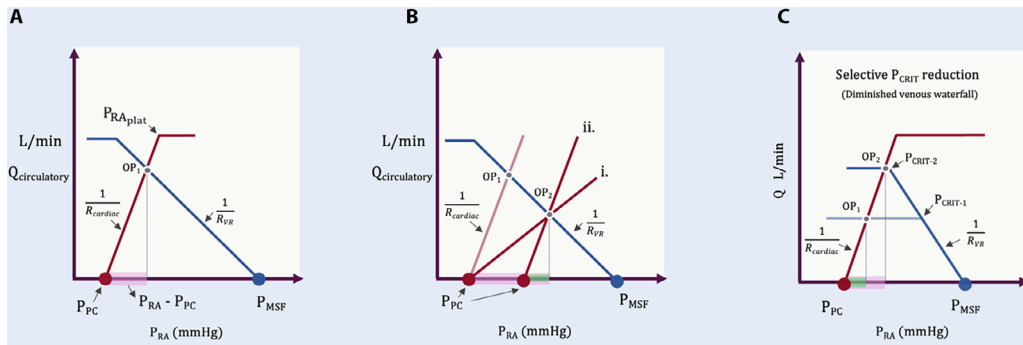


Figure 2. The Geometrical Model and the three types of venous congestion. A.) Shows the approximation of VExUS distending pressure as the pink bar as the base of the cardiac triangle (hypotenuse is $1/R_{\text{CARDIAC}}$). The pink bar on the x-axis represents P_{RA} minus P_{PC} or the transmural P_{RA} . The intersection of cardiac function (red) and venous return curve (blue) forms the operating point 1 (OP_1) which defines the P_{RA} and total blood flow (Q). B.) The difference between increased transmural P_{RA} (i.) versus increased absolute P_{RA} (ii.). The increased transmural pressure (pink bar, x-axis) is from impaired cardiac function (i.e., increased R_{CARDIAC}) while the decreased transmural (green bar, x-axis) but increased absolute P_{RA} (ii.) is caused by elevated P_{PC} . Both i. and ii. have the same operating point 2 (OP_2) compared to baseline (OP_1). R_{CARDIAC} plateau is not shown for figure clarity. C.) Shows the effect of increased P_{CRIT-1} (e.g., from tumor burden) at operating point 1 (OP_1); the initial VExUS is illustrated as the green bar on the x-axis. Reduction of tumor burden decreases P_{CRIT} , increases Q and P_{RA} (OP_2) with diminished organ congestion and an increase in VExUS illustrated as the pink bar on the x-axis. P_{MSF} is mean systemic filling pressure and for discussion purposes considered the same as P_{MSA} .

List of Abbreviations

F_{TOL}: fluid tolerance
 GM: Geometrical Model
 VExUS: venous excess ultrasound score
 P_{MSA}: mean systemic pressure analogue
 F_R: volume responsive
 F_{UR}: volume unresponsive
 R_{VR}: resistance to venous return
 R_{CARDIAC}: cardiac resistance or cardiac function
 P_{RA}: right atrial pressure
 P_{PC}: pericardial pressure
 P_{RA-plat}: right atrial pressure when cardiac function curve forms plateau (flattens)
 P_{CRIT}: critical collapse pressure
 OP: operating point
 MAP: mean arterial pressure
 BSA: body surface area
 E_A:E_{ES}: arterial elastance to end-systolic elastance

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